

Multi-Organ Foundation Model for Universal Ultrasound Image Segmentation With Task Prompt and Anatomical Prior

Haobo Chen, Yehua Cai, Changyan Wang, Lin Chen, Bo Zhang,
Hong Han, Yuqing Guo, Hong Ding[✉], and Qi Zhang[✉]

Abstract—Semantic segmentation of ultrasound (US) images with deep learning has played a crucial role in computer-aided disease screening, diagnosis and prognosis. However, due to the scarcity of US images and small field of view, resulting segmentation models are tailored for a specific single organ and may lack robustness, overlooking correlations among anatomical structures of multiple organs. To address these challenges, we propose the Multi-Organ FOUndation (MOFO) model for universal US image segmentation. The MOFO is optimized jointly from multiple organs across various anatomical regions to overcome the data scarcity and explore correlations between multiple organs. The MOFO extracts organ-invariant representations from US images. Simultaneously, the task prompt is employed to refine organ-specific representations for segmentation predictions. Moreover, the anatomical prior is incorporated to enhance the consistency of the anatomical structures. A multi-organ US database with segmentation labels, comprising 7039 images from 10 organs across various regions of the human body, has been established to develop and evaluate our model. Results demonstrate that the MOFO outperforms single-organ methods in terms of the Dice coefficient, 95% Hausdorff distance and average symmetric surface distance with statistically sufficient margins.

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Haobo Chen, Changyan Wang, and Qi Zhang are with the Smart Medicine and AI-Based Radiology Technology (SMART) Laboratory, Shanghai Institute for Advanced Communication and Data Science, School of Communication and Information Engineering, Shanghai University, Shanghai 200444, China (e-mail: haobo.chen@shu.edu.cn; wangcy@shu.edu.cn; zhangq@t.shu.edu.cn).

Yehua Cai and Hong Ding are with the Department of Ultrasound, Huashan Hospital, Fudan University, Shanghai 200040, China (e-mail: sunnei@msn.com; ding_hong@fudan.edu.cn).

Lin Chen and Yuqing Guo are with the Department of Ultrasound, Huadong Hospital, Fudan University, Shanghai 200040, China (e-mail: linchen11@fudan.edu.cn; gyq_hd2022@163.com).

Bo Zhang is with the Department of Ultrasound in Medicine, East Hospital, Tongji University School of Medicine, Shanghai 200120, China (e-mail: zhangbo2016@tongji.edu.cn).

Hong Han is with the Department of Ultrasound, Zhongshan Hospital, Fudan University, Shanghai 200032, China (e-mail: han.hong@zs-hospital.sh.cn).

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Our experiments in multi-organ universal segmentation for US images serve as a pioneering exploration of improving segmentation performance by leveraging semantic and anatomical relationships within US images of multiple organs.

Index Terms—Ultrasound image, segmentation, multi-organ, foundation model, task prompt, anatomical prior.

I. INTRODUCTION

SEGMENTATION of ultrasound (US) images is an essential component of medical image analysis, integral to informing clinical decision-making [1], [2]. It allows clinicians to outline tissues, improving the identification of lesions and anatomical features [3]. However, the blurred tissue boundaries and complex speckle noise in US images pose challenges for manual delineation, leading to high intra- and inter-observer variability, prompting the need for automated segmentation [4].

Deep learning has achieved remarkable success in addressing the segmentation task of US images, largely attributed to the manual annotations provided by experts [5], [6]. However, constraints such as time, expense and clinical usage limit the number of available US images. Furthermore, the high cost of pixel-level segmentation annotations contributes to a scarcity of annotated data for segmentation, often limited to a small dataset (e.g., several hundred) with expert annotations. The scarcity of data severely restricts model development, raising concerns about the robustness of models constructed based on such limited datasets.

Compared with other imaging modalities such as CT and MRI, the US imaging often presents images with a smaller and focused field of view, where only a single organ is shown in an image [7], [8]. Here, we attempt to transit from single-organ segmentation to multi-organ universal segmentation, by fully utilizing different US image datasets collected from various organs to mitigate the impact of limited samples on model development. However, the substantial differences of anatomical structures among various organs pose new challenges for establishing a multi-organ universal segmentation model. To address this issue, we collect datasets from 10 organs in various human body parts with diverse anatomical structures, as illustrated in Fig. 1, to develop a multi-organ segmentation

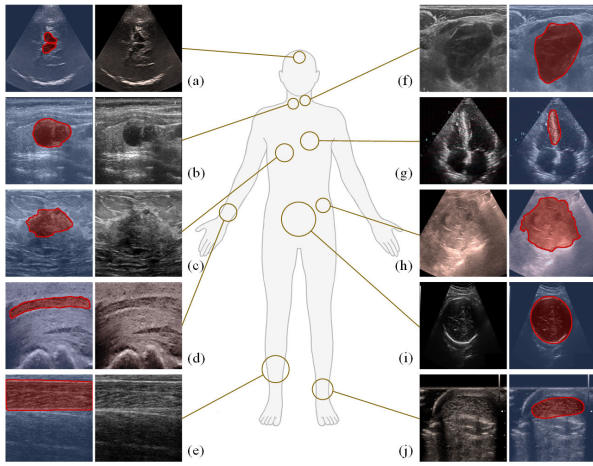


Fig. 1. Ultrasound images of various organs in the human body. (a) Midbrain. (b) Thyroid. (c) Breast. (d) Median nerve. (e) Leg muscle. (f) Lymph node. (g) Heart. (h) Kidney. (i) Fetus. (j) Achilles tendon.

foundation model, capable to learn both organ-invariant and organ-specific representations from various organs.

The organ-invariant representation refers to the characteristics that exhibit consistency and universality across various organs, unaffected by appearance of specific organs, *e.g.*, the speckle noise patterns, as well as the tissue edges represented by the acoustic impedance differences between the object of interest and background. The segmentation foundation model comprehensively understands images and effectively adapts to different anatomical structures by capturing organ-invariant representations. Moreover, it serves to mitigate the impact of variations in US imaging quality arising from differences in imaging devices and procedures. For instance, there are variations in imaging views using different probes such as convex probe (*e.g.*, Fig. 1a, 1h and 1i), sector probe (*e.g.*, Fig. 1g), and linear probe (remaining figures in Fig. 1).

In contrast to the emphasis on consistency in organ-invariant representations, organ-specific representations focus on capturing the unique characteristics of each organ to enhance the model's awareness of the heterogeneity among various organs. Learning organ-specific representations enables the model to comprehend and differentiate the anatomical differences across various organs, such as the mass-like objects including the thyroid nodule (Fig. 1b), breast tumor (Fig. 1c), lymph node (Fig. 1f), and kidney tumor (Fig. 1h), and longitudinal tubular objects including the leg muscle (Fig. 1e) and median nerve (Fig. 1d).

Existing studies typically employ U-shaped networks for direct segmentation of US images depicting a single specific organ [5], [9], [10]. Single U-shaped networks fail to account for potential anatomical correlations between organs and therefore are not suitable for multi-organ segmentation due to their inability to optimize specific anatomical characteristics across different organs.

In this paper, the **Multi-Organ FOundation (MOFO)** model with the vision, prompt, and prior branches is proposed and optimized on an US image segmentation database featuring 10 organs. Unlike previous segmentation approaches focusing on single-specific organs, the MOFO simultaneously learns

from multiple segmentation tasks related to different organs. It benefits from robust organ-invariant and organ-specific representations. Our primary contributions are summarized as follows:

- 1) Our MOFO is a segmentation foundation model formalized for multi-organ US universal segmentation, composed of the vision, prompt and prior branches, where the vision branch directly extracts organ-invariant representations from US images.
- 2) The task prompt is generated by text embedding of the multi-organ task description in the prompt branch, and aggregated with image features to refine organ-specific representations of anatomical structures.
- 3) A multi-organ auto-encoder is developed to construct a multi-organ anatomical prior space. Then its encoder serves as the anatomical prior encoder in the prior branch to map the anatomical structures of various organs into the prior space and apply constraints for the MOFO.
- 4) A hybrid US image database consisting of diverse images of 10 organs from patients across multiple medical centers is assembled to demonstrate the effectiveness of the MOFO.

II. RELATED WORK

A. Multi-Organ Segmentation

Multi-organ segmentation tasks entail labeling distinct organ regions in images [11]. CT and MRI offer a wide imaging field capable of encapsulating multiple organs, and thus deep learning on CT and MRI, with its robust representation capabilities, could exploit contextual information among multiple organs to achieve good segmentation performance. Fang and Yan [12] proposed a multi-scale deep neural network with the pyramid convolution for multi-organ segmentation in CT images. Boutillon et al. [13] developed a segmentation model using shared convolutional kernels and domain-specific normalization on MRI of three musculoskeletal joints. Shen et al. [14] proposed a spatial attention block to highlight various organs of interest in abdominal CT segmentation by learning spatial attention maps. Zhao et al. [15] combined a CNN-Transformer model and a progressive sampling module to perform multi-organ segmentation on CT and MRI images. Ulrich et al. [16] leveraged multiple CT datasets and the correlation in annotated data to train a model for various structure segmentation.

However, US imaging presents inherent constraints such as limited imaging field of view, resulting in US images featuring solitary organs. These constraints impede the multi-organ segmentation tasks on US images. Here, we collect US datasets of various individual organs and establish a foundation model for multi-organ universal segmentation on US images.

B. Task Prompt for Segmentation

The task prompt aims to enhance model awareness of key characteristics by designing effective prompts or guiding information, thereby improving segmentation performance and generalization. In contrast to traditional segmentation models

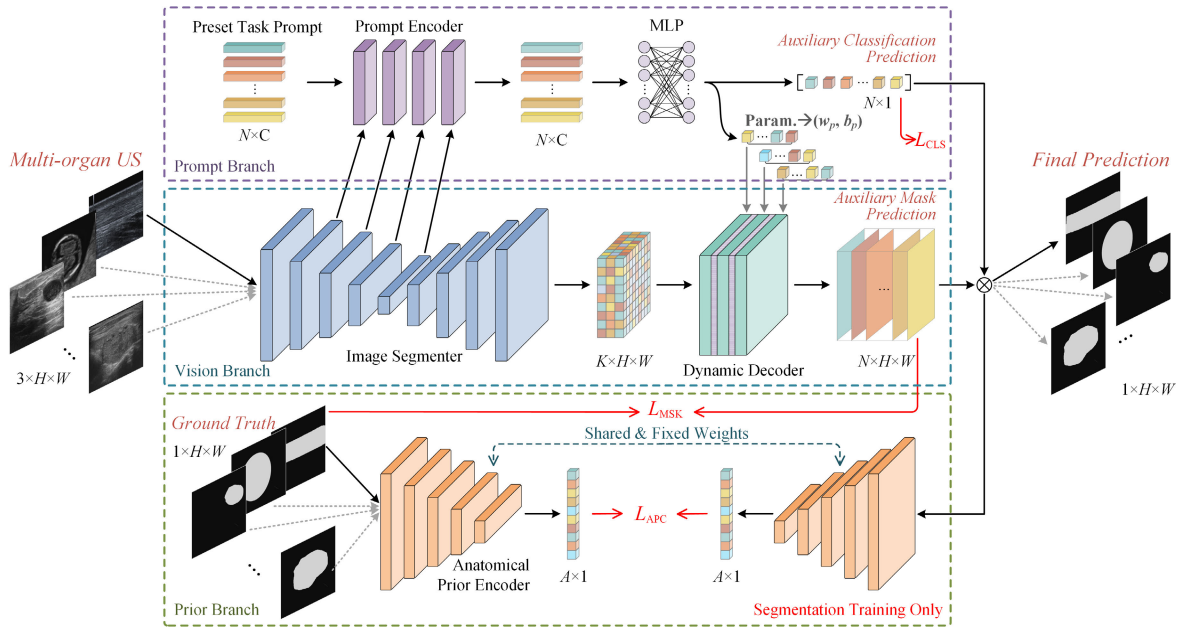


Fig. 2. Overview of the proposed MOFO model for segmenting multi-organ ultrasound images. The MOFO comprises prompt, vision, and prior branches. The vision branch extracts organ-invariant representations from images for segmentation. The prompt branch aggregates task prompt and image features to generate organ-specific representations, guiding the vision branch in segmenting objects. The prior branch utilizes the anatomical prior encoder to map the prediction maps and ground truth into the prior space, applying constraints to ensure consistency.

only utilizing single U-shaped backbones, the task prompt often adds extra branches to learn prompt information to guide and enhance the semantic segmentation. Zhang et al. [17] utilized a one-hot encoded task prompt to communicate the expected segmentation objects to the multi-organ model. However, a non-learnable one-hot task prompt is insufficient, ignoring the internal correlations between various organs. MaskFormer [18] introduces learnable queries as the task prompt, allowing them to automatically explore prompt information during the learning process. Recently, the learnable task prompts have been applied in the medical domain, underlining the importance of semantic relationship between anatomical structures in segmentation. Qin et al. [19] utilized pre-trained vision language model for image comprehension and visual question answering system to generate flexible medical prompts in the segmentation task. Liu et al. [20] established a universal model, incorporating text embedding from the contrastive language-image pre-training (CLIP) to capture anatomical relationships to segment multiple organs and tumors. This paper endeavors to design a learnable task prompt to explore organ-specific representations between various organs.

C. Prior Learning

Prior learning is a regularization technique in deep learning that constrains the model by incorporating prior knowledge or assumptions, aiming to improve generalization ability and mitigate overfitting risks. It typically introduces additional penalty terms to the loss function, imposing constraints related to shape, size, topology, or other correlations among different objects in segmentation [21], [22]. However, the diversity in organ morphology makes it impractical to manually design a

universal prior knowledge applicable for each organ in multi-organ segmentation, necessitating a more adaptable approach.

A deep auto-encoder emerges as a promising solution, serving as an effective shape discriminator for prior knowledge. By leveraging self-supervised learning strategies, it can autonomously acquire prior knowledge without relying on manually defined rules. Chen et al. [23] utilized an auto-encoder to learn shape priors for refining segmentation tasks. Zhou et al. [24] employed auto-encoders to learn self-supervised representations, combined with supervised representations to enhance diagnostic capability. Boutillon et al. [25] applied a deep auto-encoder to learn priors from anatomical structures of ground truth. Here, we further expand a deep auto-encoder into our task, aiming to achieve multi-organ anatomical prior knowledge to capture the characteristics of anatomical structures from various organs.

III. METHODS

A. Overall Framework

The MOFO model comprises a prompt branch, a vision branch and a prior branch, as illustrated in Fig. 2. Initially, the preset task prompt is introduced to describe various organs and aggregate image features through the prompt encoder, yielding organ-specific representations in the prompt branch. Simultaneously, a U-shaped image segmenter is employed to extract global organ-invariant representations from multi-organ US images in the vision branch. Subsequently, the generated representations are utilized to guide the dynamic decoder to generate the auxiliary classification and mask prediction and get the final prediction of the segmentation mask. Furthermore, an anatomical prior encoder with fixed weights is used to extract anatomical structure representations from

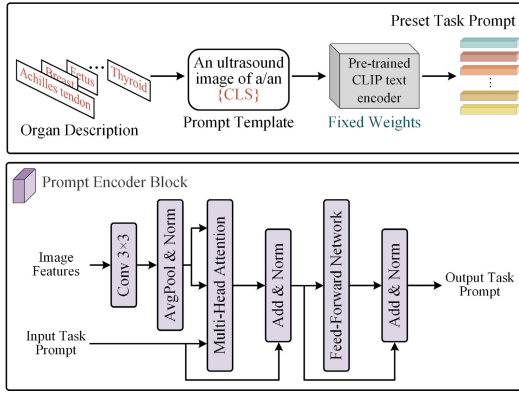


Fig. 3. Details of the preset task prompt and prompt encoder block.

final prediction and ground truth, respectively, thus further constraining the output in the prior branch during the training stage.

B. Prompt Branch

1) **Preset Task Prompt:** The task prompt is crucial for MOFO to generate organ-specific representations for multi-organ segmentation. With the widespread success of large pre-trained visual-language models in the medical domain, we attempt to leverage CLIP to design a task prompt that understands the semantic relationships among anatomical structures of different organs [6], [26].

Referring to the text branch of CLIP, each organ description is successively inserted into a prompt template and then fed into the pre-trained text encoder to generate the prompt embedding $u_k \in \mathbb{R}^{1 \times L}$ associated with the k -th class organ, where L denotes the length of the prompt embedding. To ensure that the prompt embeddings match our segmentation task, we design the prompt template as “An ultrasound image of a/an {CLS}”, where {CLS} represents the name of an organ, e.g., breast. All the prompt embeddings are aggregated into an embedding group $u_G = \{u_1, u_2, \dots, u_N\}$, where N denotes the number of organs. Given that the fixed u_G is prone to semantic bias, we input it into a multi-layer perceptron (MLP) to generate a learnable task prompt, i.e., $u_T = \text{MLP}(u_G)$, which serves as the preset task prompt in Figs. 2 and 3.

2) **Prompt Encoder:** The prompt branch utilizes the prompt encoder (PE) to aggregate image features and the preset task prompt, generating organ-specific representations. The prompt encoder comprises four groups of encoder blocks, which are evenly distributed across feature maps of varying resolutions. The task prompt u_T is combined with image features to produce the organ-specific representations, which are then inputted into an MLP to generate auxiliary classification prediction P_{CLS} and organ-specific guiding parameters (w_p, b_p) , i.e., $P_{CLS}, (w_p, b_p) = \text{MLP}(\text{PE}(u_T, f^k))$, where $P_{CLS} \in \mathbb{R}^{N \times 1}$, and f^k denotes the image feature maps associated with the k -th organ. The auxiliary classification prediction reinforces the capacity to discriminate among distinct organs, and (w_p, b_p) is employed to guide the mask prediction in the vision branch.

The prompt encoder block is architecturally structured (Fig. 3). By utilizing the attention mechanism, it effectively combines image features and the task prompt, bolstering anatomical information. Considering that the image features from the image segmenter are redundant, and different from the dimensions suggested by the task prompt, we use convolution and pooling operations to compress the image features. Thus the prompt encoder block is defined as:

$$\tilde{f}^l = \text{AvgPool}(\text{Conv}(f^l)) \quad (1)$$

$$\tilde{u}^{l-1} = \text{MHA}(u^{l-1}, \tilde{f}^l) + u^{l-1} \quad (2)$$

$$u^l = \text{FFN}(\tilde{u}^{l-1}) + \tilde{u}^{l-1} \quad (3)$$

where u^l and f^l denote the output of the l -th prompt encoder block and the corresponding image feature, respectively. AvgPool and Conv denote the average pooling and convolution operation, respectively. MHA and FFN denote multi-head attention and feed-forward network, respectively.

C. Vision Branch

1) **Image Segmenter:** The image segmenter adopts the U-shaped structure, consisting of the Cross-Shaped Window (CSWin) Transformer-based encoder branch and CNN-based decoder branch. The encoder branch comprises five stages of CSWin Transformer blocks, followed immediately by the decoder branch, which includes four stages of convolution blocks. A 3×3 convolution-based block is set for down-sampling or up-sampling between stages. The weights pre-trained on ImageNet are applied to the encoder of the image segmenter.

2) **Dynamic Decoder:** The image features, extracted by the image segmenter, encapsulate anatomical information. The dynamic decoder is proposed in combination with the guiding parameters (w_p, b_p) from the prompt branch, to reinforce organ-specific anatomical information for the auxiliary mask prediction P_{MSK} (Fig. 2). Specifically, the essence of the dynamic encoder is the custom convolution. We use w_p and b_p as parameters for convolution kernel and bias, respectively, to participate in convolution operations (CO), i.e., $F^{c+1} = \text{CO}(F^c, w_p, b_p)$, where F^c denotes the feature maps of the c -th convolution operation. For $P_{MSK} \in \mathbb{R}^{N \times H \times W}$, individual prediction channels are established for each organ to provide abundant anatomical structure information. Ultimately, the final prediction for the organ segmentation is generated through a multiplication operation, combined with auxiliary mask prediction and classification prediction.

D. Prior Branch & Multi-Organ Auto-Encoder

A multi-organ auto-encoder is proposed to automatically learn the anatomical structure priors of various organs. The proposed auto-encoder adopts a fully convolutional U-shaped framework, as shown in Fig. 4. Additionally, the model integrates global average pooling operation to extract multi-scale representations of anatomical structure. All structure representations are combined in pairs, with representations from the same organ as positive pairs and representations from different organs as negative pairs. A multi-scale consistency learning

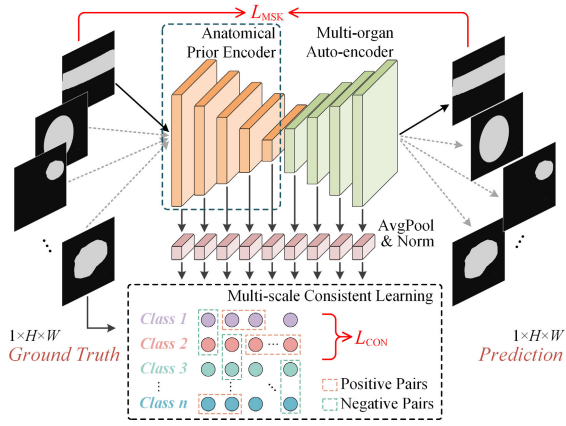


Fig. 4. Detailed structure of multi-organ auto-encoder.

strategy is employed to improve the similarity between positive sample pairs to narrow the intra-class distance and reduce the similarity between negative sample pairs to expand the inter-class distance at each scale, so as to build an anatomical prior space for various organs.

Subsequently, the encoder of the multi-organ auto-encoder serves as an anatomical prior encoder (Fig. 2), which is integrated into MOFO in the prior branch. It should be noted that the weights of the anatomical prior encoder remain frozen to project mask prediction and ground truth into the same anatomical space, yielding two anatomical representations. Finally, the anatomical prior constraints on the segmentation objects are enforced by computing the similarity between these two representations.

E. Loss Function

Multiple hybrid loss functions are developed to train our MOFO model. For the auxiliary classification prediction, the weighted cross-entropy loss L_{CLS} is utilized to mitigate the imbalanced class distribution:

$$L_{CLS} = - \sum_{k=1}^N v_k y_k \log(p_k) \quad (4)$$

where $y_k \in \{0, 1\}$ and $p_k \in [0, 1]$ denote the ground truth and predicted probability of the k -th class organ, respectively. v_k denotes the weight assigned to the k -th class organ, which is the reciprocal of the number of the organ.

For the auxiliary mask prediction, L_{MSK} consists of Dice coefficient loss and cross-entropy loss. Due to the independent channels assigned by the dynamic decoder for each organ, numerous all-zero ineffective channels may arise in predictions. To address this issue, we mask the loss terms associated with organ classes not present in ground truth and selectively backpropagate accurate supervision to update MOFO. Thus L_{MSK} is defined as:

$$L_{MSK} = 1 - \frac{2 \times \sum_{m=1}^{H \times W} y_m^k p_m^k}{\sum_{m=1}^{H \times W} y_m^k + \sum_{m=1}^{H \times W} p_m^k} - \sum_{m=1}^{H \times W} (y_m^k \log(p_m^k) + (1 - y_m^k) \log(1 - p_m^k)) \quad (5)$$

where $y_m^k \in \{0, 1\}$ and $p_m^k \in [0, 1]$ denote the ground truth and predicted probability of the m -th pixel belonging to the k -th class organ. H and W denote the height and width of the segmentation mask, respectively.

For the anatomical structure representation from the anatomical prior encoder, we introduce an anatomical prior consistency loss (L_{APC}) using the Euclidean distance:

$$L_{APC} = \sqrt{\sum_{a=1}^A (\mu_a - \varphi_a)^2} \quad (6)$$

where $\mu_a, \varphi_a \in [0, 1]$ denote the anatomical structure representation of the a -th pixel from the mask prediction and ground truth, and A denotes the length of the structure representation.

MOFO is ultimately trained using three above loss functions to construct a hybrid loss function L_{SEG} , as follow,

$$L_{SEG} = L_{MSK} + \lambda_1 L_{CLS} + \lambda_2 L_{APC} \quad (7)$$

where λ_1 and λ_2 are weighting factors.

It should be noted that before training the MOFO, the multi-organ auto-encoder is utilized to obtain the anatomical prior. L_{MSK} is used to constrain the anatomical structure of each organ, along with a consistency loss (L_{CON}) for multi-scale anatomical structure representations. Let $\delta_s^{x,k}$ and $\delta_s^{y,k}$ be the structure representations of the k -th organ at scale s in one batch with n^B samples, where $x, y \in [1, \dots, n^k]$ ($x \neq y$), $s \in [1, \dots, n^s]$. n^k and n^s denote the number of the k -th organ in one batch and the number of the structure representation scale, respectively. Meanwhile, let $\delta_s^{x',k'}$ be the structure representation of the k' -th organ at scale s , and k' denotes other organs except k in the same batch, where $x' \in [1, \dots, n^B - n^k]$. L_{CON} is defined as follows:

$$L_{CON} = - \frac{1}{n \times n^s} \sum_{s=1}^{n^s} \sum_{k=1}^N \sum_{x=1}^{n^k} \frac{1}{n^k} \sum_{y=1, x \neq y}^{n^k} \log \frac{\exp(\varepsilon(\delta_s^{x,k}, \delta_s^{y,k}))}{\sum_{x'=1}^{n^B - n^k} \exp(\varepsilon(\delta_s^{x,k}, \delta_s^{x',k'}))} \quad (8)$$

where ε denotes the cosine similarity operation. Hence, the loss of the multi-organ auto-encoder becomes

$$L_{AE} = L_{MSK} + \lambda_3 L_{CON} \quad (9)$$

with an empirically set weighting factor λ_3 .

IV. EXPERIMENTS

A. Datasets for Training, Validation and Test

To fully explore the potential of the proposed MOFO model in multi-organ universal segmentation, we assemble an US database comprising 7039 images, encompassing 10 organs or tissues from different parts of the human body. It consists of four publicly available datasets and six private datasets. The detail of the ultrasound database is presented in Table I.

The breast dataset [27], fetus dataset [28], leg muscle dataset [29] and thyroid dataset [30] are open datasets, including 2592 images. The Achilles tendon dataset, median nerve

TABLE I

DETAIL OF THE ULTRASOUND DATABASE INCLUDING 10 ORGANS

Source	Organ	People	Image	Target
Open	Breast	-	647	Breast tumor
	Fetus	-	999	Fetal head
	Leg muscle	-	309	Lower leg muscle
	Thyroid	-	637	Thyroid nodule
Private	Achilles tendon	172	1048	Achilles tendon
	Heart	218	218	Interventricular septum
	Kidney	392	1102	Kidney tumor
	Lymph node	815	815	Lymph node
	Median nerve	520	1059	Median nerve
	Midbrain	205	205	Midbrain

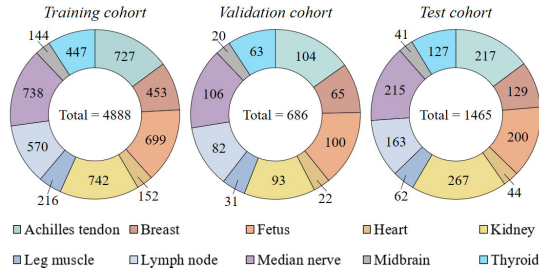


Fig. 5. Statistics of the multi-organ ultrasound image database.

dataset and midbrain dataset were collected from Huashan Hospital, Fudan University, Shanghai. The heart dataset was acquired from East Hospital, Tongji University, Shanghai. The kidney dataset was collected from Huadong Hospital, Fudan University, Shanghai. The lymph node dataset originated from Zhongshan Hospital, Fudan University, Shanghai. Our six private datasets consist of 4447 images from 2226 diseased people or people without disabilities in total.

The multi-organ US image database is divided into three cohorts based on the person level, with a ratio of 7:1:2, for the training, validation and test cohorts. The distribution of US images for each organ across the cohorts is illustrated in Fig. 5. Multiple images from the same person are allocated to a single cohort to prevent data leakage.

B. Datasets for Out-Of-Distribution Evaluation

To evaluate the out-of-distribution (OOD) performance of MOFO, we assemble two OOD datasets on the breast and liver. The breast dataset [31] is publicly available and contains 232 US images with breast tumors, and the medical center and its imaging equipment and settings are different from those of the breast dataset [27] included in the multi-organ database. The liver dataset includes 256 US images with liver tumors from Huashan Hospital, Fudan University, Shanghai.

C. Implementation Details

Our proposed MOFO is implemented using Pytorch 2.0.0. The training and inference of the MOFO are conducted using four RTX3090 cards, with a batch size of 80 and image size of $3 \times 224 \times 224$. Models are trained in the training cohort and then the best-performance model is selected based on the validation cohort, which finally is used to calculate

TABLE II

DETAILS OF THE KEY ARCHITECTURE IN MOFO

Description	Variable	Dimension
Height of image	H	224
Width of image	W	224
Number of organs	N	10
Length of prompt embedding	L	256
Channel of feature map from Image Segmenter	K	32
Length of anatomical prior representations	A	256

metrics in the test cohort. Furthermore, the Dice similarity coefficient (DSC), 95% Hausdorff distance (HD95) and average symmetric surface distance (ASSD) are used to evaluate the segmentation performance. In MOFO, CLIP employs ViT-B/32 as the backbone, using its pre-trained text encoder to convert organ descriptions into the task prompt. Details of the key architecture in MOFO are shown in Table II. The weighting factors of the loss function are empirically set as $\lambda_1 = 0.5$, $\lambda_2 = 0.2$ and $\lambda_3 = 0.5$.

V. RESULTS

We compare the performance of our proposed MOFO with each individual model trained on each single-organ dataset. We quantify the contribution of modules proposed in MOFO through ablation studies. Moreover, we validate the impact of various preset task prompts of the prompt branch, image segmenters of the vision branch and anatomical prior encoders of the prior branch in MOFO.

A. Comparison With An Individual Model Trained on A Single Organ

To ensure a fair comparison, an individual model is constructed by adopting the image segmenter as the backbone and adding a convolution layer to transform the prediction from $K \times H \times W$ to $1 \times H \times W$. The auto-encoder is optimized using a single-organ dataset to provide simple anatomical priors for the individual models.

Table III presents the segmentation performance comparison of single-organ individual models and the MOFO. The MOFO significantly outperforms individual models across all 10 organs (all $p < 0.05$, most < 0.001). Compared with individual models, the MOFO improves the segmentation performance of various organs differently. The MOFO shows highly significant metric improvement for objects like the kidney tumor and lymph node, which have blurred and indistinct contours. For objects with clearer contours, such as the fetal head, leg muscle and median nerve, the improvement of the MOFO in metrics is modest but still significant.

The segmentation visualization of above models is presented in Fig. 6. The MOFO approximates the boundaries of the target objects. Compared with individual models, the MOFO exhibits fewer false-positive and false-negative regions. For the heart in Sample C, the lymph node in Sample A and the median nerve in Sample B, the MOFO effectively avoids misidentification of background areas with similar echogenicity to the target region. For the kidney tumor with the ambiguous edge, although individual models can feasibly locate the target

TABLE III
SEGMENTATION PERFORMANCE OF INDIVIDUAL MODELS AND THE MOFO

	DSC (%)			HD95 (pixel)			ASSD (pixel)		
	Individual	MOFO	<i>p</i> -value	Individual	MOFO	<i>p</i> -value	Individual	MOFO	<i>p</i> -value
Achilles tendon	84.69	87.49	<0.001	20.56	15.02	<0.001	7.03	5.64	<0.001
Breast	82.17	88.18	<0.001	19.17	13.64	<0.001	5.98	4.01	<0.001
Fetus	96.96	97.47	<0.001	4.77	3.93	<0.001	1.91	1.57	<0.001
Heart	82.68	87.30	<0.001	10.97	8.04	0.016	3.10	2.18	<0.001
Kidney	70.23	76.73	<0.001	21.70	18.36	<0.001	7.68	6.61	<0.001
Leg muscle	96.23	97.93	<0.001	14.05	4.02	<0.001	2.74	1.24	<0.001
Lymph node	79.61	82.89	0.001	30.96	24.63	0.005	9.37	7.97	<0.001
Median nerve	88.72	91.42	<0.001	10.83	7.04	<0.001	2.92	2.12	<0.001
Midbrain	85.35	89.75	0.002	13.40	4.82	<0.001	3.19	1.56	<0.001
Thyroid	79.88	83.29	0.007	22.53	19.20	0.041	7.86	6.84	0.035
Average	84.65	88.25	<0.001	16.89	11.87	<0.001	5.18	3.97	<0.001

P-values are calculated with the Wilcoxon signed rank test between two models.

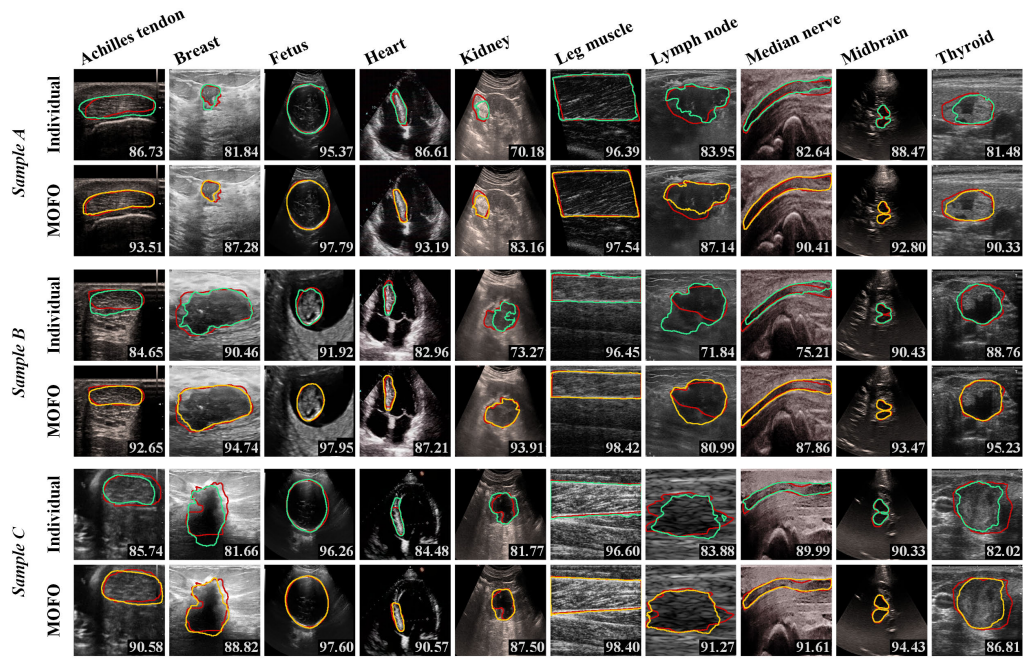


Fig. 6. Visualization of segmentation results of typical samples using individual models and the MOFO. The ground truths by manual delineations are depicted in red, and the predictions of individual models and the MOFO are depicted in green and orange, respectively. The value in the lower right of each image is the DSC value.

position, there is a significant deviation in contour localization. The MOFO learns from segmentation tasks of other organs, leading to improved contour localization for kidney tumor segmentation. Notably, for the midbrain segmentation, where its anatomical structure resembles a butterfly shape with distinct boundaries between two wing-like regions, individual models exhibit vague recognition of boundary. In contrast, the MOFO effectively recognizes the boundary area and distinguishes the midbrain into two distinct parts.

B. Performance on Objects With Different Shapes

Morphological assessment is crucial for disease diagnosis. To further explore the advantages of the MOFO, morphological features are analyzed for the segmentation results of individual models and the MOFO, including area, hull area, perimeter, center distance, major axis, minor axis, equivalent diameter, rotation angle, circularity, aspect

ratio, solidity and eccentricity. The root mean square error (RMSE) of each morphological feature is calculated between the ground truth and prediction to assess their consistency, as shown in Fig. 7. The MOFO exhibits smaller RMSEs, demonstrating its superiority over individual models in capturing diverse anatomical structures and morphological features.

Fig. 8 offers a comprehensive comparison of the performance on mass-like objects including the breast tumor, kidney tumor, lymph node and thyroid nodule with irregular shapes between the ground truth and predictions, through Bland-Altman plots. A detailed analysis of three morphological features, including circularity, centroid distance, and convex area, is conducted. Compared to individual models, the MOFO demonstrates narrower consistency limits for these three features across four organs, indicating robust segmentation performance, stability and consistency.

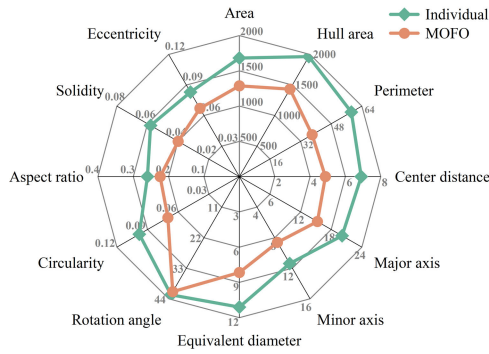


Fig. 7. Comparison of RMSE for morphological features between the ground truth and prediction.

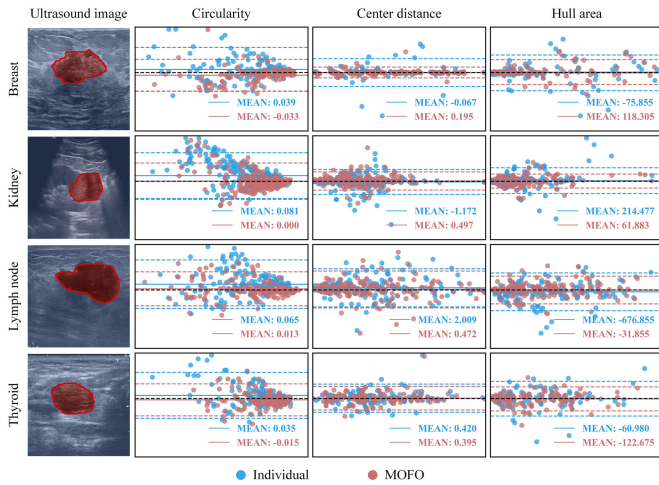


Fig. 8. Bland-Altman plots of morphological features for mass-like objects including the breast tumor, kidney tumor, lymph node and thyroid nodule. In each plot, the horizontal axis denotes the mean value derived from model prediction and ground truth, while the vertical axis denotes the difference value between model prediction and ground truth. The solid lines delineate the mean difference values and the dashed lines delineate the limits of the 95% consistency limits.

C. Comparison With Multi-Organ State-Of-The-Art Models

To evaluate the performance of MOFO, six multi-organ state-of-the-art (SOTA) models are compared with our MOFO, including the SAM [32], MedSAM [6], SAM-Med2D [33], AdaptiveSAM [34], SAMUS [35] and UniverSeg [36]. The SAM is evaluated using the automatic segmentation and the interactive segmentation with point and bounding box (Bbox) as prompts. The MedSAM is an interactive segmentation using point, Bbox and text as prompts. The SAM-Med2D utilizes point and Bbox as prompts. The UniverSeg randomly selects 64 ultrasound images and the corresponding ground truth from the training cohort as the support input for segmentation. Table IV presents the segmentation performance of the MOFO compared with four multi-organ SOTA models. The MOFO exhibits superior performance on most organs, achieving an average DSC of 88.25% across ten organs. Benefiting from training on lots of ultrasound images, the MedSAM using Bbox prompts achieves the second-best overall segmentation performance. Interestingly, for the segmentation of kidney

tumor, lymph node and thyroid nodule, the MedSAM with Bbox prompts outperforms the MOFO in DSC, though not showing superior performance in HD95 and ASSD. It should be noted that here the MedSAM uses Bbox as prompts, and the restriction of the segmentation area into the pre-set Bbox has made the segmentation task relatively easier. It is unfair to compare the MedSAM with our MOFO, which automatically locates the target within the entire image without a pre-set Bbox. The SAM, designed for natural images, performs worse than the MedSAM. The SAM-Med2D with only a limited amount of US images in its training set, also shows relatively poor performance. The SAMUS, which uses points as prompts and is specifically designed for ultrasound images, does not demonstrate particularly outstanding segmentation performance across the 10 organs. The AdaptiveSAM, using text as prompts, shows a slightly lower average DSC compared to SAMUS. However, it is noteworthy that AdaptiveSAM achieves a DSC of 82.71% in segmenting the midbrain, ranking second only after MOFO. Although the UniverSeg uses support input from corresponding organ datasets in segmentation, its performance is not satisfactory.

The segmentation visualization of the above models is presented in Fig. 9. The MOFO achieves the best segmentation performance, most closely approximating the ground truth. For the samples of the Achilles tendon and median nerve, the MOFO can delineate the object contours smoothly and clearly, whereas other methods produced rough and uneven contours. For the midbrain segmentation, the MOFO effectively identifies the butterfly-shaped anatomical structure, while other methods can only locate mass-like objects, which do not accurately reflect the structure of the midbrain. The MedSAM and SAM both with Bbox show similar performance, while the performance of UniverSeg is lower than other methods, especially in object localization. The UniverSeg often identifies multiple incorrect targets for the breast, kidney and median nerve in Sample A and the heart, kidney and thyroid in Sample B.

D. Comparison With Single-Organ State-Of-The-Art Models

In addition to the aforementioned four multi-organ SOTA models, five single-organ SOTA models are included in our comparison, including the U-Net [10], UNet++ [37], nnU-Net [9], TransUNet [38] and Swin-Unet [39]. Each model is trained and evaluated on one dataset specific to a single organ, and the results are averaged across all 10 organs. Table V presents the segmentation performance of MOFO compared with the five single-organ SOTA models. The MOFO outperforms five single-organ models across all 10 organs, achieving the best segmentation performance with the average DSC of 88.25%, HD95 of 11.87 pixels and ASSD of 3.97 pixels. The Swin-Unet achieves sub-optimal performance. Compared to the worst-performing U-Net, the MOFO results in a 6.01% increase in terms of the DSC, and 10.62 pixels and 2.51 pixels decrease in terms of the HD95 and ASSD.

Furthermore, each of the five single-organ models is trained and evaluated together on all 10 organs, and is compared with the MOFO. Fig. 10 presents the segmentation performance

TABLE IV
DICE SIMILARITY COEFFICIENTS (DSCs, %) OF OUR MOFO AND SIX MULTI-ORGAN SOTA MODELS

	SAM			MedSAM			SAM-Med2D		AdaptiveSAM	SAMUS	UniverSeg	MOFO
	-	Point	Bbox	Point	Bbox	Text	Point	Bbox	Text	Point		
Achilles tendon	2.70	42.06	85.87	46.38	85.96	1.21	7.13	25.03	75.38	76.20	74.40	87.49
Breast	16.89	68.99	87.44	69.06	86.38	22.17	80.60	85.92	73.30	84.89	65.98	88.18
Fetus	9.37	66.20	94.28	74.30	95.70	1.08	8.15	19.48	92.54	94.03	90.40	97.47
Heart	0.00	19.90	80.46	37.88	76.07	0.94	28.62	43.82	81.83	70.26	71.40	87.30
Kidney	0.71	30.82	76.29	53.47	78.90	16.05	44.36	59.21	43.90	68.22	38.16	76.73
Leg muscle	44.82	70.21	91.53	44.17	89.23	2.67	3.48	22.33	94.56	90.72	95.19	97.93
Lymph node	16.38	61.79	81.68	70.46	83.98	3.76	53.92	74.62	75.07	75.07	71.38	82.89
Median nerve	1.72	33.21	68.24	14.94	69.97	0.40	35.39	56.09	77.31	78.13	77.26	91.42
Midbrain	0.00	11.34	74.62	39.49	76.79	2.47	20.80	44.74	82.71	63.27	80.70	89.75
Thyroid	2.58	34.71	80.11	66.96	86.15	20.31	31.93	65.56	65.48	80.94	62.61	83.29
Average	9.52	43.92	82.05	51.71	82.91	7.11	31.44	49.68	76.21	78.17	72.75	88.25



Fig. 9. Visualization of segmentation results of typical samples using the SAM with Bbox, MedSAM with Bbox, UniverSeg and MOFO. The ground truths by manual delineations are depicted in red, and the predictions of the MOFO and other comparison models are depicted in orange and green, respectively. The value in the lower right of each image is the DSC value.

TABLE V
SEGMENTATION PERFORMANCE OF THE MOFO AND FIVE SINGLE-ORGAN MODELS, EACH OF WHICH IS TRAINED AND EVALUATED ON ONE DATASET SPECIFIC TO A SINGLE ORGAN

	DSC (%)	HD95 (pixel)	ASSD (pixel)
U-Net	82.24±7.40	22.49±8.55	6.48±2.68
UNet++	84.41±7.75	17.78±7.24	5.39±2.56
nnU-Net	85.36±7.29	16.36±7.76	5.00±2.54
TransUNet	84.38±7.93	17.11±9.54	5.38±3.12
Swin-Unet	85.52±7.62	13.48±7.13	4.65±2.68
MOFO	88.25±6.49	11.87±7.33	3.97±2.57

of the MOFO compared with five comparison models. The MOFO outperforms all five comparison models across all

10 organs, achieving superior segmentation performance with the highest DSC and the lowest HD95 and ASSD. Additionally, the nnU-Net delivers sub-optimal performance, while the U-Net performs the worst.

E. Ablation Studies

The MOFO comprises the vision, prompt and prior branches. To validate the effectiveness of these branches in MOFO, we design models with different configurations for ablation studies. In the vision branch, the image segmenter utilizes a U-shaped network with CSWin Transformer as the encoder and CNN as the decoder, as shown in Table VI. On this basis, we introduce the prompt branch to guide the

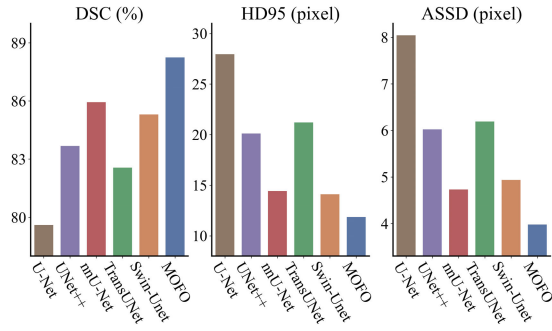


Fig. 10. Segmentation performance of the MOFO and five single-organ models, each of which is trained and evaluated together on all 10 organs.

TABLE VI
ABLATION STUDY FOR THREE BRANCHES IN THE MOFO

Vision branch	✓	✓	✓
Prompt branch	×	✓	✓
Prior branch	×	×	✓
Achilles tendon	84.82 19.10 6.84	87.21 15.06 5.65	87.49 15.02 5.64
Breast	81.23 19.62 6.31	85.91 14.39 4.50	88.18 13.64 4.01
Fetus	96.67 5.48 2.07	97.09 4.36 1.73	97.47 3.93 1.57
Heart	84.42 9.74 2.74	86.41 8.69 2.33	87.30 8.04 2.18
Kidney	68.93 24.35 8.47	73.48 17.78 6.94	76.73 18.36 6.61
Leg muscle	96.22 6.76 1.93	97.88 4.28 1.27	97.93 4.02 1.24
Lymph node	81.48 27.08 8.35	81.01 24.84 8.45	82.89 24.63 7.97
Median nerve	89.05 11.48 2.90	90.90 8.28 2.31	91.42 7.04 2.12
Midbrain	87.56 6.64 2.02	88.43 5.37 1.65	89.75 4.82 1.56
Thyroid	77.89 25.19 8.65	80.80 21.91 7.69	83.29 19.20 6.84
Total average	84.83 15.54 5.03	86.91 12.50 4.25	88.25 11.87 3.97

The indices are denoted in a form of “DSC (%) | HD95 (pixel) | ASSD (pixel)”.

vision branch in segmenting objects. This results in a significant improvement of 2.08% in DSC, along with reductions of 3.01 pixels in HD95 and 0.78 pixels in ASSD. Subsequently, the introduction of the prior branch further constrains the anatomical structure of the output objects, leading to the complete MOFO model, which exhibits the best segmentation accuracy. It is noteworthy that the MOFO exhibits optimal performance across all organ segmentation tasks.

F. Comparison of Different Preset Task Prompts

To explore the segmentation performance of the preset task prompt of the MOFO, seven different preset task prompts are designed for comparison, including (a) random parameter embedding, (b) one-hot class embedding, (c) BioBERT [40] prompt with the template “An ultrasound image of a/an {CLS}”, (d) BioLinkBERT [41] prompt with the template “An ultrasound image of a/an {CLS}”, (e) CLIP prompt with the template “{CLS}” (CLIP v1), (f) CLIP prompt with the template “A photo of a/an {CLS}” (CLIP v2), and (g) CLIP prompt with the template “An ultrasound image of a/an {CLS}” (CLIP v3), where CLS represents an organ name.

The segmentation performance of the MOFO using seven preset task prompts is shown in Table VII. The results indicate that the segmentation performance using CLIP-based prompts or BERT-based prompts is significantly superior to that using random parameters or one-hot embedding as the task prompt. The task prompts constructed with random or one-hot embedding ignore the inter-organ correlations. Compared to the

simple template (CLIP v1) and the original CLIP template (CLIP v2), the template of CLIP v3 is specifically optimized for our US segmentation tasks. The MOFO using CLIP v3 achieves the optimal DSC, HD95 and ASSD. The DSC shows an increase of 1.27%, accompanied by a decrease of 1.37 pixels and 0.4 pixels in HD95 and ASSD, respectively, compared to MOFO using random parameters as the task prompt. The MOFO using the BioLinkBERT achieved suboptimal results, with performance similar to the MOFO using CLIP v2.

G. Comparison of Different Image Segmenters

To assess the influence of various image segmenters in the MOFO, apart from the CSWin Transformer-based image segmenter, we introduce image segmenters utilizing ResNet [42], EfficientNet [43], PoolFormer [44] and Swin Transformer [45] as their encoder, respectively. As shown in Table VIII, the MOFO utilizing CSWin Transformer achieves the best performance with the DSC of 88.25%, HD95 of 11.87 pixels and ASSD of 3.97 pixels. The MOFO using EfficientNet achieves sub-optimal segmentation performance. The MOFO utilizing ResNet shows the worst performance, decreasing by 2.03% compared to the proposed MOFO using CSWin Transformer.

H. Comparison of Different Prior Auto-Encoders

To evaluate the anatomical prior encoder, the multi-organ auto-encoder is trained with various loss functions. Detailed configurations are as follows: (a) L_{MSK} solely imposes mask constraints, (b) L_{CLS} incorporates class constraints on multi-scale structure representations in addition to L_{MSK} , and (c) L_{CON} utilizes the multi-scale consistency learning to enforce constraints on the multi-scale structure representations, combined with L_{MSK} (our final method).

The anatomical space is constructed utilizing the multi-organ auto-encoder trained with three loss functions, as depicted in Fig. 11. A disordered anatomical space is established by solely imposing mask constraint (L_{MSK}), where the anatomical information is intertwined in Fig. 11a. The addition of class constraints (L_{CLS}) to L_{MSK} improves the confusion of anatomical information, especially for objects with unique anatomical structures like the fetus, heart, leg muscle, median nerve, and midbrain in Fig. 11b. However, the effectiveness of L_{CLS} is limited, and the confusion persists between the mass-like objects. To address this limitation, our proposed method replaces L_{CLS} with multi-scale consistency constraint (L_{CON}) to reinforce consistency constraints on multi-scale structure representations. It effectively groups together the same organs and enhances the anatomical boundaries between various organs in Fig. 11c.

After building the multi-organ auto-encoder, its encoder serves as the anatomical prior encoder of the MOFO. The segmentation performance of the MOFO employing various anatomical prior encoders is presented in Table IX. The MOFO utilizing $L_{MSK} + L_{CON}$ to construct the anatomical space effectively facilitates multi-organ segmentation, yielding the best segmentation performance. Compared to not using the anatomical prior encoder, it results in a 1.33% increase in DSC, while HD95 and ASSD decrease by 0.63 pixels

TABLE VII
SEGMENTATION PERFORMANCE OF THE MOFO WITH SEVEN PRESET TASK PROMPTS

Embedding	Prompt template	DSC (%)	HD95 (pixel)	ASSD (pixel)
Random	-	86.98±7.18	13.24±7.97	4.34±2.73
One-hot	-	87.05±7.59	12.28±7.74	4.16±2.73
BioBERT	An ultrasound image of a/an {CLS}	87.21±6.83	12.71±6.84	4.21±2.47
BioLinkBERT	An ultrasound image of a/an {CLS}	87.65±6.63	12.00±7.02	4.10±2.59
CLIP v1	{CLS}	87.57±6.79	12.43±7.50	4.11±2.61
CLIP v2	A photo of a/an {CLS}	87.64±6.75	12.40±7.65	4.10±2.62
CLIP v3	An ultrasound image of a/an {CLS}	88.25±6.49	11.87±7.33	3.97±2.57

TABLE VIII
SEGMENTATION PERFORMANCE OF THE MOFO WITH FIVE ENCODERS FOR IMAGE SEGMENTER

	DSC (%)	HD95 (pixel)	ASSD (pixel)
ResNet	86.21±7.76	14.05±8.37	4.62±2.90
EfficientNet	87.16±7.42	12.36±7.28	4.21±2.65
PoolFormer	86.77±7.29	13.95±7.59	4.44±2.64
Swin Transformer	86.92±6.83	14.23±7.55	4.47±2.54
CSWin Transformer	88.25±6.49	11.87±7.33	3.97±2.57

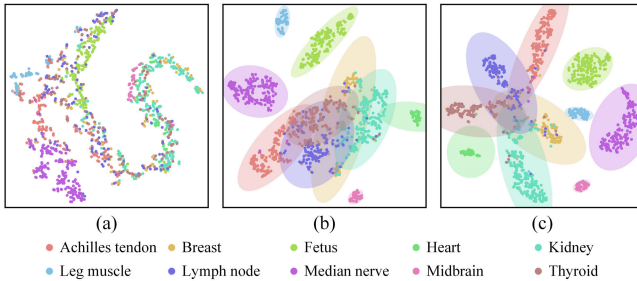


Fig. 11. T-SNE visualization of the anatomical spaces constructed three loss functions. (a) is trained with L_{MSK} . (b) is trained with $L_{MSK} + L_{CLS}$. (c) is trained with $L_{MSK} + L_{CON}$.

TABLE IX
SEGMENTATION PERFORMANCE OF THE MOFO WITH DIFFERENT PRIOR AUTO-ENCODERS

	DSC (%)	HD95 (pixel)	ASSD (pixel)
None	86.91±7.42	12.50±7.42	4.25±2.75
L_{MSK}	86.50±7.32	13.12±7.22	4.49±2.66
$L_{MSK} + L_{CLS}$	87.59±6.49	13.40±7.31	4.15±2.62
$L_{MSK} + L_{CON}$	88.25±6.49	11.87±7.33	3.97±2.57

and 0.28 pixels, respectively. Additionally, the MOFO utilizing $L_{MSK} + L_{CLS}$ achieves suboptimal performance. It is noteworthy that the MOFO solely utilizing L_{MSK} shows the worst performance. We suspect that the confused multi-organ anatomical prior space disrupts the latent correlations between various organs, thereby affecting the segmentation performance.

I. Out-Of-Distribution Performance

To evaluate the OOD performance of MOFO, the MedSAM using Bbox as prompts, is fine-tuned on our multi-organ US database and is compared with the MOFO. As shown in Fig. 12a, the fine-tuned MedSAM and MOFO achieve comparable performance in terms of the DSC. It should be noted that the MedSAM benefits from the restriction of the segmentation area to the pre-set Bbox that interactively contributes to

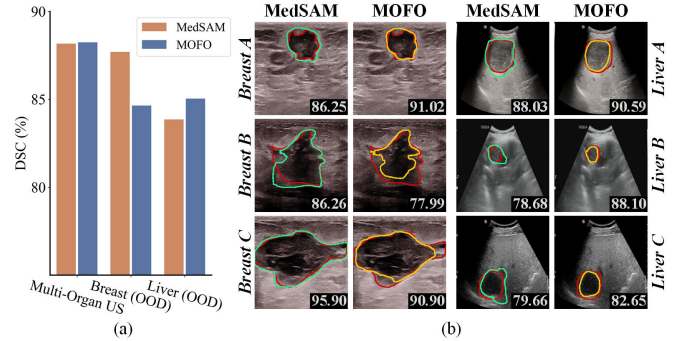


Fig. 12. Segmentation performance of our MOFO and the MedSAM. (a) Dice similarity coefficients (DSCs, %) on the multi-organ US database and the out-of-distribution (OOD) datasets of breast and liver. (b) Visualization of segmentation results of OOD samples. The ground truths by manual delineations are depicted in red, and the predictions of MOFO and MedSAM are depicted in orange and green, respectively. The value in the lower right of each image is the DSC value.

the segmentation task. In contrast, the MOFO automatically locates targets across the entire image, achieving performance comparable to the interactive MedSAM on the OOD breast dataset. Moreover, the MOFO outperforms MedSAM in terms of segmentation accuracy on the OOD liver dataset.

The visualization of the OOD segmentation results for both models is presented in Fig. 12b. Both the MOFO and MedSAM accurately localize tumors and achieve high DSC-values. However, they exhibit different segmentation tendencies. The MOFO tends to produce slightly smaller tumor segmentations compared to the ground truth, whereas the MedSAM often obtains slightly larger tumor regions.

VI. DISCUSSION

A. Innovative Aspects of MOFO

A multi-organ foundation model named MOFO is proposed for universal US segmentation. Experiments conducted on an US image database, comprising 10 different organs, demonstrate its advanced performance. The architecture of the MOFO consists of three branches. In the vision branch, the image segmenter adopts a U-shaped model based on CSWin Transformer to generate organ-invariant representations. In the prompt branch, the task prompt is a text embedding, generated with the pre-trained CLIP and a manually designed prompt template for quantification of the correlations between multiple-organ segmentation tasks. The prompt encoder enhances organ-specific representations by learning the task prompt and image features. This assists

the dynamic decoder in the vision branch in utilizing organ-specific representations derived from the prompt branch to generate the segmentation predictions. The prior branch is active only during the training phase. It projects the prediction maps and ground truth into a latent anatomical prior space and introduces anatomical structure constraints, thereby leading to improve segmentation performance.

B. MOFO & Other Foundation Models: A Comparison

There are two types of foundation models, the task-agnostic and the task-specific models. The recent USFM is a pioneering foundational model tailored for US images, functioning as a task-agnostic foundation model [46]. However, the USFM is trained using unsupervised learning, which does not require pixel-level expert annotations, resulting in lacking inherent practical applications and necessitating fine-tuning downstream tasks to adapt to specific requirements. In contrast, our MOFO is a task-specific foundation model designed for US image segmentation, using supervised learning. By leveraging a large and diverse set of US images from our US image database, the MOFO explores the relationships and distinctions between multiple organs, achieving superior segmentation performance. It is therefore noteworthy that our MOFO stands as one of the first basic US image models and the first basic model specifically designed for US image segmentation.

The SAM and its variants, as the task-specific foundation models, have demonstrated exceptional performance in segmentation, and the prompts, *e.g.*, point, Bbox and text significantly enhance the model's segmentation accuracy [6], [32], [33], [34], [47]. The SAMs using these prompts are classified as an interactive segmentation approach, necessitating user input to guide and refine the segmentation process for precise outcomes. The efficacy of the SAMs is highly dependent on the accuracy of user-provided prompts. However, it is particularly challenging and time-intensive to deliver precise input prompts in medical image segmentation. Our MOFO is a fully automatic segmentation foundation model, only requiring the input of the US image to be segmented, without any additional input of the specific prompts. The MOFO automatically matches text embeddings of specific organs mentioned in the task prompt based on input US images.

C. Assessment of Representations for Task Prompt

Appropriate task prompt presets, incorporating descriptions of various organs and tasks, can enhance the model's awareness of different anatomical structures, thereby improving segmentation performance. The MOFO leverages the robust text understanding capability of CLIP to generate the task prompt, indirectly indicating a correlation in medical interpretation between various organs (Fig. 13). The breast, kidney, lymph node and thyroid, which exhibit morphological similarities in US images, show high squared correlations. The high correlation between the Achilles tendon and leg muscle is evident, which aligns with their physiological and anatomical relationship as components of the human motion system [48]. As an extension of the lower leg muscles, the Achilles tendon

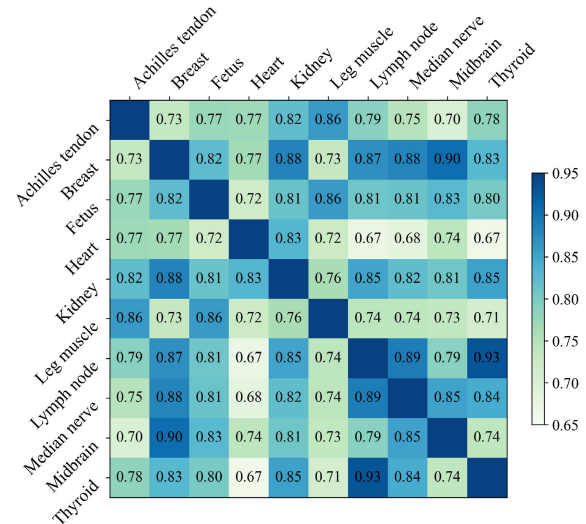


Fig. 13. Squared correlations between 10 organs in the task prompt.

transmits the force generated by the triceps surae to the heel, thereby facilitating plantar flexion of the foot [49], [50]. Interestingly, despite lacking a direct and close anatomical or physiological connection, some organs may have relatively high correlations, such as between the median nerve and lymph node, and between the breast and midbrain. Furthermore, the MOFO leverages the task prompt from the organ descriptions, enhancing the perception of the MOFO for the anatomical structures of various organs through the semantic relationships between them.

D. Benefits for Clinical Practice

Different from MRI and CT, which provide wider field of view that shows multiple organs in a single image, in the US imaging, there are usually only one specific organ of interest in a single image. Obtaining a sufficient number of finely annotated US images for a specific organ is often challenging, hindering the construction of deep learning models. Meanwhile, the correlations between various organs are neglected in the modeling of single-organ segmentation for US images. To address these issues, the MOFO is proposed to establish a robust segmentation model, leveraging multiple organ datasets. An US image database is assembled to train the MOFO, comprising 7039 images from various organs, populations, and centers, including both normal and diseased organs/tissues with segmentation masks. The MOFO proves that the foundation model relying on multi-organ data can improve the performance of universal multi-organ segmentation. By effectively utilizing existing imaging resources, the MOFO outperforms individual models trained solely on single-organ data.

The targets segmented by the MOFO involve 10 organs, some are the organs themselves, while others are diseased tissues within organs. For example, the targets in the breast dataset and kidney dataset are the breast tumors and kidney tumors rather than the whole breast or whole kidney, the same goes to the thyroid dataset, where the target is the thyroid nodule within the thyroid, rather than the whole thyroid.

Of course, the breast tumor, kidney tumor and thyroid nodule are abnormal tissues with “abnormal shapes”. Therefore, the MOFO is well-suited for segmenting organs/tissues with normal shapes as well as those with deformed shapes due to disease. Morphological analysis reveals that the anatomical regions segmented by the MOFO align more closely with radiologist-annotated anatomical regions in terms of shapes and contours. It reduces the overall cost of acquiring medical images and achieves satisfactory performance, making it a potentially practical solution to computer-aided diagnosis.

E. Limitations and Perspectives

We have collected a total of 7039 US images with detailed annotations from both public and private datasets, but these datasets are limited to 10 organs, that means our model by far can only cope with universal segmentation of these 10 organs. There is a pressing need to expand the categories and numbers of organs in the future. During data collection, we observed that the scarcity of data is mainly due to the laborious and time-consuming image annotation. Unlabeled data is relatively easier to obtain compared to labeled data, and thus we plan to incorporate unlabeled data and explore semi-supervised learning strategies in the future updated version of the MOFO.

In addition, we have quantified the correlations among multiple organs and attempted to enhance the interpretability of the correlations through Figs. 11 and 13. However, the weights learned by neural networks lack intuitive interpretability. An attempt is being made to introduce a large-scale image-text model to directly generate textual descriptions of the correlations between multiple organs, aiming to increase confidence in model decisions and thereby improve the reliability of US image analysis.

VII. CONCLUSION

We propose a segmentation foundation model, the MOFO, for the universal segmentation of multiple organs in US images. The MOFO consists of the vision, prompt, and prior branches. The vision branch extracts organ-invariant representations and reconstructs segmentation masks. The prompt branch combines image features with the task prompt for various organs to guide the vision branch to segment objects. The prior branch imposes anatomical structure constraints on the output masks, improving the segmentation performance. The MOFO adapts to 10 various organs and achieves superior segmentation performance. The MOFO serves as an innovative exploration to improve universal segmentation by employing the task prompt and anatomical relationships of multiple organs.

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